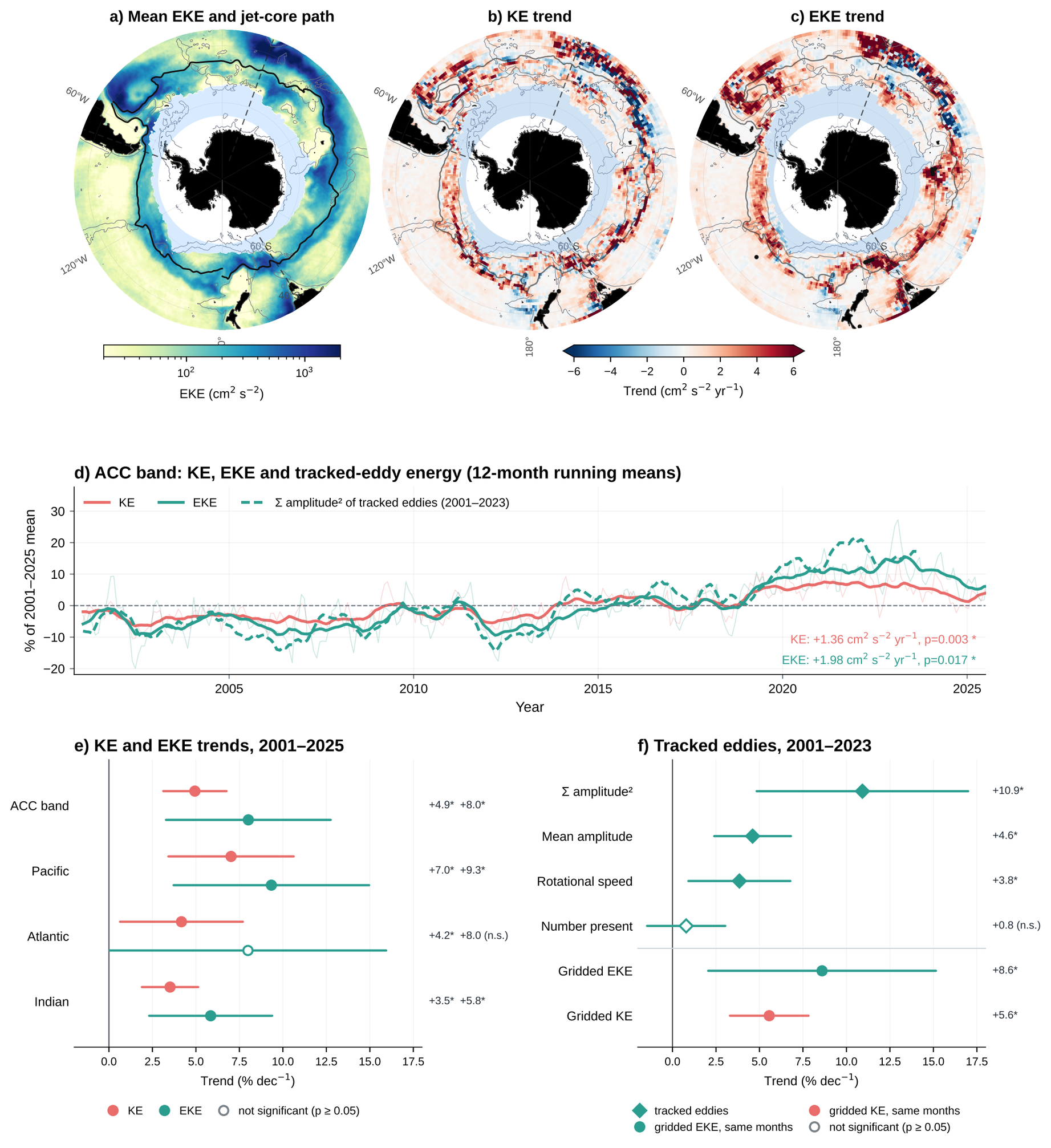

a) Mean EKE and jet-core path
b) KE trend
c) EKE trend
60°W
120°W
60°S
40
0°
180°
10^2
10^3
EKE (cm^2 s^{-2})
−6 −4 −2 0 2 4 6
Trend (cm^2 s^{-2} yr^{-1})
d) ACC band: KE, EKE and tracked-eddy energy (12-month running means)
KE
EKE
Σ amplitude² of tracked eddies (2001–2023)
% of 2001–2025 mean
30
20
10
0
−10
−20
2005
2010
2015
2020
2025
Year
KE: +1.36 cm^2 s^{-2} yr^{-1}, p=0.003 *
EKE: +1.98 cm^2 s^{-2} yr^{-1}, p=0.017 *
e) KE and EKE trends, 2001–2025
ACC band
Pacific
Atlantic
Indian
+4.9* +8.0*
+7.0* +9.3*
+4.2* +8.0 (n.s.)
+3.5* +5.8*
0.0 2.5 5.0 7.5 10.0 12.5 15.0 17.5
Trend (% dec^{-1})
KE
EKE
not significant (p ≥ 0.05)
f) Tracked eddies, 2001–2023
Σ amplitude²
Mean amplitude
Rotational speed
Number present
Gridded EKE
Gridded KE
+10.9*
+4.6*
+3.8*
+0.8 (n.s.)
+8.6*
+5.6*
0.0 2.5 5.0 7.5 10.0 12.5 15.0 17.5
Trend (% dec^{-1})
tracked eddies
gridded EKE, same months
gridded KE, same months
not significant (p ≥ 0.05)